Transformer-enhanced multi-agent reinforcement learning for cooperative encirclement of unmanned surface vessels

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ABSTRACT

Cooperative multi-target encirclement by Unmanned Surface Vessel (USV) swarms poses a challenging problem in maritime operations, due to the increasing difficulty of maintaining effective coordination as swarm size grows. To address this challenge, this paper proposes a novel Transformer-enhanced multi-agent proximal policy optimization (TMAPPO) framework. Specifically, we design a policy network that integrates a Transformer encoder to focus on critical spatial information, while a GRU module is further introduced to enhance temporal decision-making capability. In addition, we design a comprehensive reward function that jointly considers target pursuit, collision avoidance, and encirclement uniformity. Meanwhile, a more realistic training environment is constructed by incorporating USV dynamic characteristics and environmental disturbances. Numerical experiments show that the proposed method achieves higher success rates and rewards than representative baselines, with its advantage becoming more pronounced as the task scale increases. Moreover, the learned policy achieves autonomous target allocation and safe multi-agent encirclement in complex scenarios.

1. Introduction

In recent years, USVs have developed rapidly and become an important component of autonomous marine systems. Benefiting from flexible deployment, relatively low operating cost, and the ability to operate in hazardous environments, USVs are being increasingly employed in environmental monitoring and maritime surveillance (Lv et al., 2025), resource exploration and offshore inspection (Liu et al., 2025), and autonomous navigation tasks (Wu et al., 2024). Cooperative multi-USV systems have attracted growing attention because they can accomplish more complex maritime tasks more efficiently and robustly through distributed collaboration (Gantiva Osorio et al., 2024; Wang and Zhao, 2025). Recent studies have investigated multi-USV cooperation in scenarios such as cooperative navigation and collision avoidance (Wang and Zhao, 2025), cooperative formation control (Lv et al., 2024; Wang et al., 2025), cooperative interception (Chen et al., 2024), and cooperative target hunting or encirclement (Zhang et al., 2025a; Xia et al., 2023). Among these tasks, cooperative encirclement has emerged as a representative decision-making problem with important applications in maritime security, target containment, and defense operations. As the numbers of both pursuers and targets increase, the interaction space grows rapidly, requiring agents to simultaneously address dynamic target assignment, precise spatial coordination, and collision avoidance in dense and highly coupled environments. Therefore, compared with single-agent missions or small-scale engagements, large-scale multi-USV multi-target scenarios are substantially more challenging.

Multi-agent reinforcement learning (MARL) has emerged as a promising paradigm for cooperative decision-making, since it enables agents to learn coordinated strategies through interaction with the environment (Yu et al., 2022; Lowe et al., 2020). Recent studies have successfully applied advanced MARL algorithms to these challenges, Zhang et al. (2025b) proposed a Proximal Policy Optimization algorithm based on situational field (SF-PPO) for dual-USV combat decision-making, in which a situational field model was incorporated into the reward design to improve convergence and generalization in adversarial scenarios. However, the method was developed in a fixed dual-USV setting with fixed-dimensional observation inputs, which limits its scalability to tasks involving dynamically changing numbers of participating agents. Similarly, Mu et al. (2026) developed an enhanced multi-agent deep deterministic policy gradient (MADDPG)-based encirclement method that explicitly incorporates reef obstacles and collision effects into multi-USV cooperative encirclement. Zhang et al. (2025a) proposed a distributed transferable MARL

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Table 1
Comparison of representative studies.

Reference	Swarm scale	Input representation	Method	Task scenario	Environment considerations
Zhang et al. (2025b)	Single-USV, single-target	Observation features	SF-PPO	Adversarial combat	dynamic model
Mu et al. (2026)	Multi-USV, single-target	Observation features	MADDPG	Cooperative encirclement	kinematic model, static obstacles
Zhang et al. (2025a)	Multi-USV, single-target	Observation features	MADDPG	Cooperative encirclement	dynamic model, static obstacles
Xia et al. (2023)	Multi-USV, multi-target	Pooled feature embedding	DPOMH-PPO with a feature embedding block	Cooperative encirclement	kinematic model
Woo and Kim (2020)	Single-USV, multi-target	Grid-map features	DDQN with CNN	Collision avoidance	dynamic obstacles

framework for cooperative target encirclement by multiple USVs, in which a centralized training with decentralized execution (CTDE) training scheme together with policy sharing was introduced to support efficient training and policy transfer across pursuer swarms of different sizes. However, both studies remain limited to single-target encirclement, which restricts their applicability to scenarios involving multiple targets. Xia et al. (2023) proposed a distributed partially observable multi-hunting proximal policy optimization (DPOMH-PPO) framework for cooperative multi-target hunting by USV fleets, where a feature embedding block combining column-wise max pooling (CMP) and column-wise average pooling (CAP) was introduced to encode partial observations. Woo and Kim (2020) proposed a double deep Q-network (DDQN) framework for USV collision avoidance, in which the encounter situation was represented as a visual grid map and processed by a convolutional neural network (CNN) for decision-making. Although these studies provide practical solutions for handling multi-entity interactions under complex observation conditions, pooling-based compression and grid-based representation inevitably lead to the loss of fine-grained inter-agent information, thereby weakening the representation of agent-specific characteristics and relative spatial relations. As a result, higher-order spatial interactions remain difficult to capture, which is unfavorable for precise encirclement strategy learning. A comparison of representative studies is provided in Table 1.

Although the aforementioned studies have advanced the development of USV encirclement, several challenges remain in large-scale multi-target encirclement. First, the number of agents involved in the task may vary dynamically, making policies trained with conventional fixed-input architectures difficult to transfer across swarm scales. Policies learned in small-scale settings usually depend on observations with fixed dimensions and therefore cannot generalize effectively to larger swarms. Second, as the swarm scale increases, the amount and complexity of interaction information grow rapidly. In large-scale multi-target encirclement, each agent must make decisions from observations involving multiple pursuers, multiple targets, and their coupled spatial relations. This makes it increasingly difficult to identify the most relevant entities and interaction patterns for decision-making, thereby hindering the learning of effective encirclement strategies. Third, more realistic training environments impose additional requirements on robustness and safety. Complex hydrodynamics, environmental disturbances, and modeling errors increase the difficulty of policy learning, while larger swarm scales further intensify collision risks and coordination constraints. As a result, policies trained under such settings must not only achieve effective encirclement, but also maintain safe and robust behavior under environmental uncertainty.

To overcome these limitations, we propose a Transformer-enhanced multi-agent proximal policy optimization (TMAPPO) framework for the multi-USV multi-target encirclement task. The main contributions of this work are summarized as follows:

1. We propose a Transformer-enhanced policy network for large-scale multi-USV multi-target encirclement. By introducing multi-head self-attention, the policy can naturally process observations with varying numbers of agents and focus on critical spatial information in complex multi-agent scenarios. A GRU module is further introduced to enhance temporal decision-making capability.

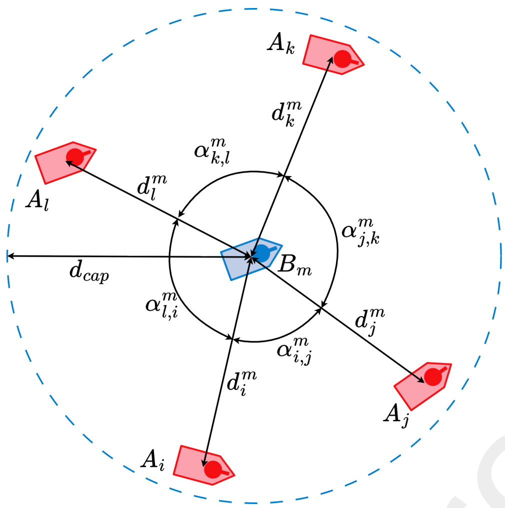

Figure 1: Geometry of single-target encirclement

2. We design a comprehensive reward function for cooperative encirclement that jointly considers target pursuit, collision avoidance, and encirclement uniformity, thereby encouraging safe and effective cooperative behaviors.
3. We construct a more realistic training environment by incorporating USV dynamic characteristics and environmental disturbances, which improves the robustness and practical relevance of the learned policy.

The rest of this paper is structured as follows. Section 2 outlines the theoretical background and the design of the USV competitive simulation platform. Section 3 provides a detailed description of the proposed TMAPPO algorithm and the training framework. The numerical results and corresponding analyses are discussed in Section 4. Some concluding remarks and suggestions for future research are given in the final section.

2. Multi-USV multi-target encirclement problem formulation

2.1. Task description

In this work, we address the multi-USV multi-target encirclement problem within a maritime environment. The scenario involves a swarm of N USVs, denoted as pursuers $\mathcal{U}^A = \{A_1, \dots, A_N\}$, and a set of M dynamic targets, denoted as evaders $\mathcal{U}^B = \{B_1, \dots, B_M\}$. The objective is to achieve encirclement of all targets through the coordinated actions of the pursuers. The geometric configuration for single-target encirclement is illustrated in Fig. 1. In the figure, d_i^m denotes the Euclidean distance between pursuer A_i and target B_m, and d_{cap} denotes the capture radius. A pursuer is regarded as actively engaging target B_m when $d_i^m \leq d_{cap}$. For each target B_m, the corresponding active pursuer subset is defined as $\mathcal{P}_m = \{A_i \in \mathcal{U}^A \mid d_i^m \leq d_{cap}\}$. The encirclement angle $\alpha_{i,j}^m$ is defined as the angular sector subtended by two pursuers A_i and A_j relative to target B_m, and α_{cap} denotes the maximum encirclement angle for successful capture. Two pursuers are defined as neighbors if no other active pursuer lies within the angular sector between them.

A target B_m is considered successfully captured if its active pursuer subset $\mathcal{P}_m$ forms a valid encirclement around it. Specifically, the encirclement must satisfy two conditions: 1) cardinality constraint, $|\mathcal{P}_m| \geq 3$; and 2) angular constraint, $\alpha_{i,j}^m \leq \alpha_{cap}$ for any two adjacent pursuers in $\mathcal{P}_m$. Once a target satisfies these capture conditions, it is deemed captured and removed from the environment. The mission terminates when all targets are captured or when the maximum time horizon is reached.

2.2. USV dynamics and environment modeling

Different from the simplified kinematic models adopted in Xia et al. (2023), each USV is modeled by a 3-Degree-of-Freedom (3-DOF) underactuated dynamic model (Skjetne et al., 2004). This 3-DOF model provides a convenient framework for explicitly incorporating environmental disturbance forces and moments into the vessel dynamics, enabling the learned policy to be trained under more realistic operating conditions. The state of each USV is defined by its position vector $\boldsymbol{\eta} = [x, y, \psi]^T$ in the earth-fixed frame and its velocity vector $\boldsymbol{\nu} = [u, v, r]^T$ in the body-fixed frame. The nonlinear dynamics are governed by:

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau} + \boldsymbol{\tau}_{env} \tag{1}$$

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\psi)\boldsymbol{\nu} \tag{2}$$

where $\mathbf{M}$ is the inertia matrix, $\mathbf{C}(\boldsymbol{\nu})$ is the Coriolis and centripetal matrix, and $\mathbf{D}(\boldsymbol{\nu})$ denotes the hydrodynamic damping matrix. The control input is $\boldsymbol{\tau} = [\tau_u, 0, \tau_r]^T$, reflecting the underactuated nature of the USV, and $\mathbf{J}(\psi)$ is the rotation matrix from the body-fixed frame to the earth-fixed frame. The hydrodynamic parameters used in this work, including the mass and damping coefficients, are selected with reference to the vessel model reported in Skjetne et al. (2004).

In this work, wave-induced loads are not modeled explicitly. This simplification is commonly adopted in decision-level USV studies using planar 3-DOF models, where the main objective is to capture the dominant drift and heading perturbations affecting cooperative maneuvering, while avoiding the substantial increase in model complexity caused by wave-frequency motions. Accordingly, the environmental disturbance term is modeled as

$$\boldsymbol{\tau}_{env} = \boldsymbol{\tau}_{wind} + \boldsymbol{\tau}_{current}, \tag{3}$$

where $\boldsymbol{\tau}_{wind}$ and $\boldsymbol{\tau}_{current}$ denote the disturbance loads induced by wind and ocean current, respectively. In this work, these disturbance loads are estimated using empirical relationships based on the environmental conditions and vessel states.

2.3. Decentralized partially observable Markov decision process (Dec-POMDP)

The cooperative multi-target encirclement problem is modeled as a decentralized partially observable Markov decision process, represented by the tuple $\mathcal{G} = \langle \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$. $\mathcal{S}$ represents the global state space, describing the information of all USVs. $\mathcal{O}$ represents the set of joint observations; due to limited sensing capabilities, each agent i only perceives a partial observation $o_{i,t} \in \mathcal{O}_i$ derived from the global state s_t. $\mathcal{A}$ is the joint action space, where each agent i selects an action $a_{i,t} \in \mathcal{A}_i$ according to its policy $\pi_i(a_{i,t}|o_{i,t})$. $\mathcal{P}(s_{t+1}|s_t, \mathbf{a}_t)$ denotes the state transition probability function governing the evolution of the environment. $\mathcal{R}(s_t, \mathbf{a}_t)$ is the shared global reward function to encourage cooperation, and $\gamma \in [0, 1)$ is the discount factor. The objective of the task is to find a joint policy $\boldsymbol{\pi}$ that maximizes the expected discounted cumulative return $J(\boldsymbol{\pi}) = \mathbb{E}[\sum_{t=0}^{T} \gamma^t r_t]$.

2.3.1. State spaces

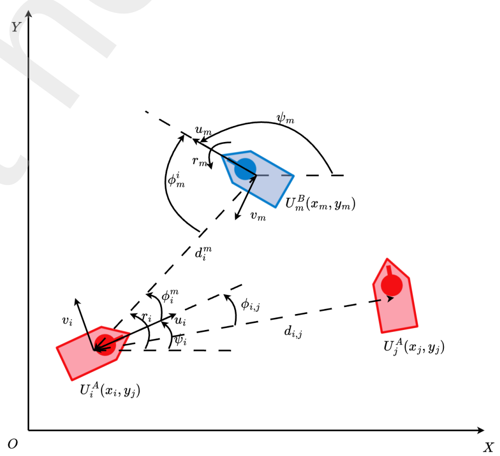

Figure 2: Schematic of relative observation features in ego-agent's body frame

To capture the global environmental dynamics, we construct a state feature vector, denoted as $\mathbf{x}_{i,t}$, for each entity in the environment at time step t. This vector consists of both the kinematic status and the semantic identity of the vessel. Specifically, the USV's pose is defined by its global coordinates (x, y) and heading ψ. To capture instantaneous motion dynamics, the state vector explicitly includes the linear and angular velocities (v_x, v_y, r) expressed in the global inertial frame. Furthermore, to differentiate agent roles within the mixed swarm, a one-hot vector is appended to the state features to strictly categorize each entity as either a pursuer or an evader.

Formally, the state vector $\mathbf{x}_{i,t}$ for the i-th USV is defined as:

$$\mathbf{x}_{i,t} = \left[x_{i,t}, y_{i,t}, \psi_{i,t}, v_{x,i,t}, v_{y,i,t}, r_{i,t}, \mathbf{c}_i\right] \tag{4}$$

where the vector $\mathbf{c}_i \in \{0, 1\}^2$ is a one-hot encoding representing the agent's faction identity:

$$\mathbf{c}_i = \begin{cases} [1, 0] & \text{if } i \in \mathcal{U}^A \text{ (Pursuers)} \\ [0, 1] & \text{if } i \in \mathcal{U}^B \text{ (Evaders)} \end{cases} \tag{5}$$

Finally, the global state of the environment at time step t, denoted as $s_t \in \mathcal{S}$, is represented as the aggregate set of the individual kinematic state vectors for all participating USVs (including both pursuers and targets):

$$s_t = \{\mathbf{x}_{1,t}, \mathbf{x}_{2,t}, \dots, \mathbf{x}_{N+M,t}\} \tag{6}$$

2.3.2. Observation spaces

While the global state s_t captures the absolute environmental state, the decision-making of each pursuer relies strictly on its local observation. Let $\mathcal{N}_i$ denote the dynamic subset of entities currently located within the limited perception range d_{sense} of the i-th pursuer. The local observation is formulated as a variable-length set of feature vectors, $o_i = \{\mathbf{o}_{i,j} \mid j \in \mathcal{N}_i\}$.

Unlike the global inertial attributes in $\mathcal{S}$, the kinematic features within each vector $\mathbf{o}_{i,j}$ are transformed into the ego-agent's body-fixed frame to facilitate local spatial reasoning. The observation vector $\mathbf{o}_{i,j}$ consists of two primary categories of features. Regarding relative kinematics, instead of using absolute coordinates, we compute geometric and dynamic attributes relative to the ego-agent. This explicitly includes the longitudinal distance Δx, lateral distance Δy, Euclidean distance $d_{i,j}$, relative bearing $\phi_{i,j}$, and heading difference $\Delta\psi$. To capture motion trends, we also include the relative linear velocities Δv_x and Δv_y alongside the relative yaw rate Δr, defined as the difference between the observed entity and the ego-agent. These geometric relationships are illustrated in Fig. 2.

Extending the global faction identity to explicitly differentiate the ego-agent, we append a relational identity vector to the kinematic features. This one-hot vector $\mathbf{c}_{i,j} \in \{0, 1\}^3$ strictly categorizes entity j as either the ego-agent itself, a teammate, or a target:

$$\mathbf{c}_{i,j} = \begin{cases} [1, 0, 0] & \text{if } j = i \text{ (Self)} \\ [0, 1, 0] & \text{if } j \in \mathcal{U}^A \setminus \{i\} \text{ (Teammate)} \\ [0, 0, 1] & \text{if } j \in \mathcal{U}^B \text{ (Target)} \end{cases} \tag{7}$$

The final observation vector is defined as:

$$\mathbf{o}_{i,j} = [\Delta x, \Delta y, d_{i,j}, \phi_{i,j}, \Delta\psi, \Delta v_x, \Delta v_y, \Delta r, \mathbf{c}_{i,j}] \tag{8}$$

To ensure numerical stability during training, all kinematic features within both s_t and o_i undergo standard normalization.

2.3.3. Action space

The design of the action space is guided by the maneuvering requirements of the task. According to the USV dynamic model introduced in the previous subsection, a USV typically controls its motion by adjusting its surge force and yaw moment. To simplify the control policy, the surge force is fixed at a constant value. Both pursuers and evaders share the same discrete action space. At each time step t, each USV i independently selects a yaw moment command $a_{i,t}$ from this space. Formally, the discrete action space is defined as:

$$\mathcal{A} = \{M_1, M_2, \dots, M_n\} \tag{9}$$

To ensure realistic physical execution, the available discrete actions are bounded by the vessel's actuation limits. Therefore, every discrete action $M_k \in \mathcal{A}$ is constrained to the feasible range of $[-1, 1]\,\mathrm{N \cdot m}$ (Zhang et al., 2025b).

2.4. Rule-based evader strategy

To provide a challenging baseline, we design a rule-based evasion strategy for the evaders using the artificial potential field (APF) method, following related studies on potential-field-based navigation for autonomous surface vessels (Jadhav et al., 2023). When pursuers enter the perception range d_{sense}, they exert a virtual repulsive force on the evader. For an evader i, the total repulsive force vector $\mathbf{F}_i = [F_{i,x}, F_{i,y}]^\top$ generated by a set of detected pursuers $\mathcal{P}_i$ is computed as:

$$\mathbf{F}_i = \begin{cases} \sum_{j\in\mathcal{P}_i} k_{rep} \left(\frac{1}{d_{i,j}} - \frac{1}{d_{sense}}\right) \frac{1}{d_{i,j}^2}\mathbf{n}_{j,i} & \text{if } \mathcal{P}_i \neq \emptyset \\ [0,0]^\top & \text{if } \mathcal{P}_i = \emptyset \end{cases} \tag{10}$$

where k_{rep} is the repulsion gain, $d_{i,j}$ is the Euclidean distance between evader i and pursuer j, and $\mathbf{n}_{j,i}$ is the unit direction vector pointing from j to i. The resultant force vector $\mathbf{F}_i$ determines the desired escape heading. This heading is tracked by a proportional-integral-derivative (PID) controller to generate a continuous yaw moment (Ye et al., 2023), and is then mapped to the nearest available value in the shared discrete action space $\mathcal{A}$.

3. Transformer-enhanced multi-agent proximal policy optimization framework

In this section, we describe the proposed framework for multi-USV cooperative encirclement. We first introduce a comprehensive reward mechanism designed to support effective encirclement while ensuring formation uniformity and navigational safety. Next, we present the neural network architectures, including the Transformer-based policy network and the centralized value network. We then describe the multi-agent proximal policy optimization (MAPPO) algorithm, including advantage estimation and policy updates. Finally, the training framework and training settings are introduced.

3.1. Reward function design

To guide cooperative encirclement, we design a comprehensive reward mechanism that jointly considers target approach, collision avoidance, and encirclement uniformity.

3.1.1. Encirclement reward

To encourage effective encirclement, we design an encirclement reward consisting of an immediate capture term and a uniformity-based dense term. When target B_m is successfully encircled by the pursuers, it is regarded as captured and removed from the environment. Simultaneously, an immediate reward is granted to each participating pursuer $A_i \in \mathcal{P}_m$ to encourage successful encirclement. The reward is defined as follows:

$$r_{i,t}^{cap} = r_{cap}, \quad \forall A_i \in \mathcal{P}_m \tag{11}$$

where r_{cap} is a predefined base capture reward.

Furthermore, to achieve successful encirclement, the pursuers should form a uniform formation around the target. To this end, we introduce a continuous shaping reward based on the angular distribution of the pursuers in $\mathcal{P}_m$. We define an encirclement uniformity score (EUS), denoted as Φ_t^{cap}:

$$\Phi_t^{cap} = \exp\left(-\sigma\left(\frac{\boldsymbol{\alpha}^m}{\bar{\alpha}}\right)\right) \tag{12}$$

where $\boldsymbol{\alpha}^m$ is the vector of adjacent angular gaps formed by the active pursuers in $\mathcal{P}_m$, $\bar{\alpha} = 2\pi/|\mathcal{P}_m|$ is the theoretical ideal gap, and $\sigma(\cdot)$ computes the standard deviation of the normalized gaps. A higher Φ_t^{cap} indicates a more evenly distributed encirclement.

Since the definition of the ideal gap $\bar{\alpha}$ shifts discontinuously when pursuers enter or leave the capture zone, directly comparing uniformity scores across varying subset sizes is mathematically invalid. Therefore, the dense shaping reward granted to each participating pursuer is conditionally formulated as:

$$r_{i,t}^{dense} = \begin{cases} \Phi_t^{cap} - \Phi_{t-1}^{cap} & \text{if } |\mathcal{P}_m(t)| = |\mathcal{P}_m(t-1)| \\ 0 & \text{otherwise} \end{cases}, \quad \forall A_i \in \mathcal{P}_m(t) \tag{13}$$

3.1.2. Target approach reward

To guide the pursuers toward the targets before encirclement, we employ potential-based reward shaping (PBRS). At each time step t, each pursuer A_i first identifies its nearest uncaptured target, denoted as B_n. We then construct a potential function $\Phi_{t,i}^{dis}$ based on the deviation from the desired capture distance d_{cap}:

$$\Phi_{t,i}^{dis} = -\left|d_i^n - d_{cap}\right| \tag{14}$$

where d_i^n denotes the Euclidean distance between pursuer A_i and its nearest target B_n.

The continuous approach reward for pursuer A_i is formulated as the temporal difference of this potential function:

$$r_i^{approach} = \gamma \Phi_{t+1,i}^{dis} - \Phi_{t,i}^{dis} \tag{15}$$

where γ is the discount factor of the RL algorithm. This difference-based mechanism provides dense feedback to guide the pursuers toward their targets and approach the desired capture distance.

3.1.3. Collision avoidance reward

To ensure encirclement safety during the pursuit, we introduce a collision penalty. Let d_i^{min} denote the minimum Euclidean distance between pursuer A_i and any other USV in the environment, including both teammates and evaders. When d_i^{min} falls below a predefined safety margin d_{coll}, the collision penalty is activated:

$$r_i^{collision} = \begin{cases} -\left(d_{coll} - d_i^{min}\right)^2, & \text{if } d_i^{min} < d_{coll} \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

The quadratic form increases the penalty rapidly as the distance between USVs falls below the safety threshold, thereby encouraging collision-avoidance behavior.

3.1.4. Total reward formulation

Ultimately, the total reward $r_{i,t}$ received by pursuer A_i at time step t is formulated as a weighted linear combination of the reward components defined above:

$$r_{i,t} = k_1 r_{i,t}^{cap} + k_2 r_{i,t}^{dense} + k_3 r_{i,t}^{approach} + k_4 r_{i,t}^{collision} \tag{17}$$

where k_1, k_2, k_3, and k_4 are scaling coefficients that balance the importance of each objective.

Furthermore, given the fully cooperative nature of the multi-USV encirclement task, it is crucial to maximize collective success rather than individual performance. To this end, we design a shared global reward mechanism. Instead of using independent individual rewards, the rewards of all active pursuers are aggregated and averaged to compute a global team reward r_t at each time step:

$$r_t = \frac{1}{N} \sum_{i=1}^{N} r_{i,t} \tag{18}$$

where N is the total number of active pursuers. Each pursuer receives this identical averaged reward r_t. This shared formulation mitigates selfish behaviors and drives the swarm toward safe and effective cooperative encirclement.

3.2. Transformer-enhanced policy architecture

The overall architecture of the proposed policy is illustrated in Fig. 3. To improve the representation of complex spatial relations among agents, we propose a Transformer-enhanced policy network. This architecture combines a Transformer encoder to focus on important spatial information with a GRU to improve temporal decision-making capability.

The local observation $o_{i,t}$ is a variable-length set of feature vectors representing the ego-agent and its neighbors. To process this unstructured input, we first employ a shared multi-layer perceptron (MLP) embedding layer to project each raw feature vector $\mathbf{o}_{i,j}$ into a d_{model}-dimensional latent space. This embedding operation generates the initial sequence of embeddings $\mathbf{E}_{i,t}^{(0)}$, enabling the network to handle a dynamic number of entities.

The core of our policy is the stack of L Transformer layers that extract critical spatial interaction information. Unlike conventional aggregation techniques that lose structural details, the multi-head self-attention (MHSA) mechanism within each layer explicitly computes the pairwise relevance between the ego-agent and its neighbors. This allows the policy to dynamically focus on the most relevant entities regardless of their number or order. Following the self-attention operation, an MLP is applied to further refine the embedding representations and enhance nonlinear feature extraction. Consequently, the output of the final layer, $\mathbf{E}_{i,t}^{(L)}$, represents a context-aware spatial representation of the swarm configuration. From this sequence, the embedding corresponding to the ego-agent, extracted as the 0-th vector, is selected for subsequent decision-making and denoted as $\mathbf{e}_{i,t} = \mathbf{E}_{i,t}^{(L)}[0]$.

To incorporate temporal dynamics, the extracted spatial feature $\mathbf{e}_{i,t} \in \mathbb{R}^{d_{model}}$ is concatenated with the ego-agent's individual kinematic state $\mathbf{x}_{i,t}$, yielding a fused feature vector $\mathbf{z}_{i,t} = [\mathbf{e}_{i,t} ; \mathbf{x}_{i,t}]$. Subsequently, $\mathbf{z}_{i,t}$ is processed by a GRU to update the temporal hidden state $h_{i,t}$. This recurrent mechanism equips the policy with historical memory and improves sequential decision-making. Finally, $h_{i,t}$ is mapped through a fully connected layer to generate action logits, followed by a softmax activation to produce the categorical action probability distribution $\pi_\theta(a_{i,t} \mid h_{i,t})$ for the current step.

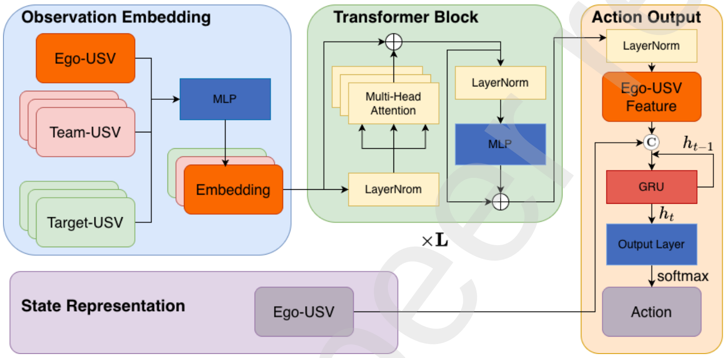

Figure 3: Transformer-enhanced policy network architecture

3.3. Centralized value network architecture

To mitigate the non-stationarity induced by independent policy updates in the multi-agent setting, we employ a centralized value network during training. Under the CTDE paradigm, this critic evaluates the cooperative strategy by estimating the joint state-value function based on the global state. The detailed architecture of this centralized critic is illustrated in Fig. 4.

Unlike the decentralized policy network, which relies on partial local observations, the centralized critic accesses the full environmental state s_t, constructed by aggregating the global feature vectors $\mathbf{x}_{i,t}$ of all entities. To accommodate the dynamically changing number of active agents within a fixed-dimensional tensor architecture, we apply a zero-masking mechanism. Specifically, when an entity becomes inactive during task execution, its global feature vector $\mathbf{x}_{i,t}$ is set to zero. This strategy maintains a constant-dimensional input matrix to handle the dynamic cardinality of the swarm, while reducing the influence of inactive entities on value estimation. The masked global state matrix is subsequently flattened and processed through an MLP to extract global state features. Parameterized by ϕ, the network ultimately outputs the estimated joint state-value $V_\phi(s_t)$.

3.4. Multi-agent proximal policy optimization

In the cooperative encirclement task, the joint state space grows exponentially with the number of USVs, and the environment exhibits strong non-stationarity from the perspective of individual agents, as their teammates' policies are constantly evolving. Standard independent policy gradient methods often fail to converge in such scenarios due to high variance in gradient estimation. To address these challenges, we adopt the MAPPO algorithm, which adheres to the CTDE paradigm.

In our framework, the policy network computes action probabilities based on local observation information to enable decentralized execution. Meanwhile, as previously defined, the centralized critic $V_\phi(s_t)$ leverages the global state s_t to estimate the joint state-value function, thereby mitigating non-stationarity during training. To estimate the

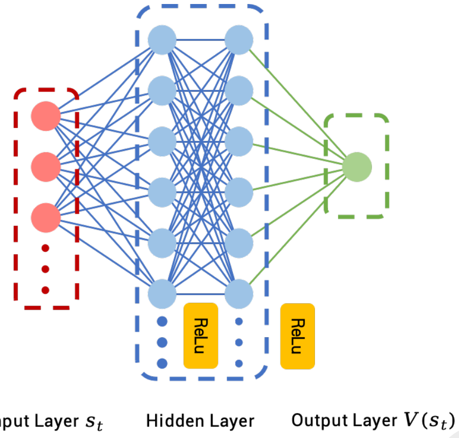

Figure 4: Centralized value network architecture

advantage function while balancing bias and variance, we employ generalized advantage estimation (GAE), which is defined at time step t as follows:

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l} \tag{19}$$

where $\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$ represents the temporal difference (TD) error of the value function, and $\lambda \in [0, 1]$ controls the bias-variance trade-off of the advantage estimation.

To promote stable policy improvement and prevent performance collapse due to excessive parameter updates, MAPPO incorporates the PPO-Clip mechanism. This restricts the magnitude of policy updates through clipping. The clipped surrogate objective limits changes that move the probability ratio $\rho_t(\theta)$ too far from unity. The clipped objective is formulated as:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min\left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right] \tag{20}$$

where $\rho_t(\theta) = \frac{\pi_\theta(a_{i,t}|h_{i,t})}{\pi_{\theta_{\text{old}}}(a_{i,t}|h_{i,t})}$ denotes the probability ratio between the new and old policies. The hyperparameter ϵ defines the clipping bandwidth and helps stabilize training. Furthermore, preventing the swarm from converging to suboptimal topologies, such as crowding on one side of the target, is important. We therefore introduce an entropy regularization term $\mathcal{H}(\pi_\theta)$ to encourage exploration and maintain diversity in action selection. Combining these components, the total loss function for the policy network is defined as

$$L_{\text{policy}}(\theta) = -L^{\text{CLIP}}(\theta) - c_{\text{ent}}\mathcal{H}(\pi_\theta) \tag{21}$$

where c_{ent} is the entropy coefficient. The centralized critic is trained to minimize the mean squared error between the predicted global value $V_\phi(s_t)$ and the return target:

$$L_{\text{critic}}(\phi) = \mathbb{E}_t \left[\left(V_\phi(s_t) - \hat{R}_t\right)^2\right] \tag{22}$$

where $\hat{R}_t = \hat{A}_t + V_\phi(s_t)$ is the computed return target.

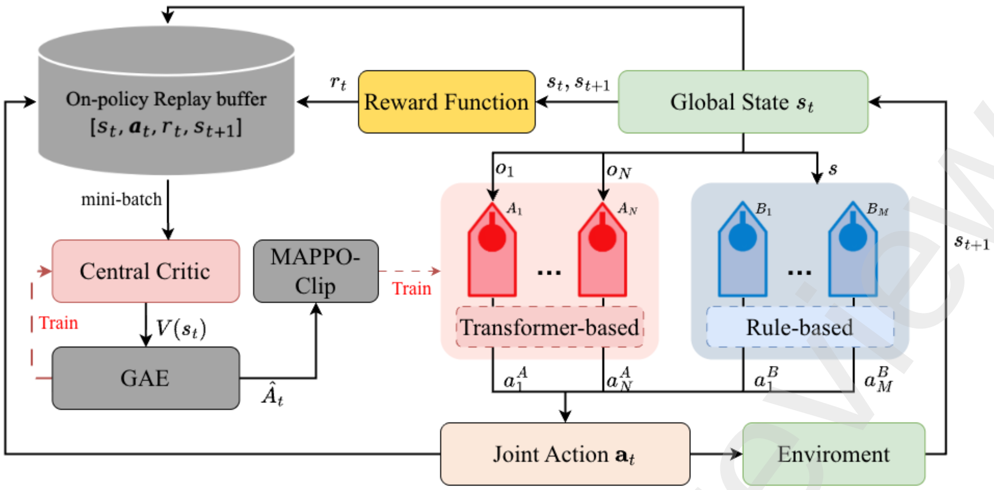

Figure 5: Training framework for multi-USV multi-target cooperative encirclement

3.5. Training framework

The overall training framework of the proposed multi-USV cooperative encirclement task is illustrated in Fig. 5, and the corresponding algorithmic execution is detailed in Algorithm 1. Initially, the algorithm inputs all hyperparameter configurations and initializes the shared policy network θ, the centralized critic network ϕ, and the on-policy rollout buffer $\mathcal{D}$.

At each time step t, each pursuer first employs the Transformer encoder to extract spatial interaction features from its current local observation. The extracted features are then integrated by the GRU to update the hidden state and generate the action from the decentralized policy π_θ. Meanwhile, each evader selects its action using the predefined APF-based strategy. After executing the joint action, the environment returns the next global state s_{t+1}, the next local observations, and the shared reward r_t. The transition data required for policy evaluation and update are stored in the rollout buffer $\mathcal{D}$. Once the number of collected rollout samples reaches the buffer capacity, the policy and value networks are updated using the collected trajectories. The detailed execution flow of the proposed framework is summarized in Algorithm 1.

3.6. Training settings

The training of the proposed multi-agent reinforcement learning framework is conducted on a deep learning workstation equipped with an Intel Core i7-13700K CPU (16 cores / 24 threads), an NVIDIA RTX 4080 Super GPU (16 GB VRAM), and 128 GB of RAM. The neural network architecture and optimization algorithms are implemented using PyTorch, while the multi-USV cooperative encirclement environment is constructed based on the standardized OpenAI Gym API to ensure structural reproducibility.

The hyperparameter configurations for the proposed algorithm are summarized in Table 2. These parameters are selected with reference to standard settings in prior MARL literature (Xia et al., 2023; Yu et al., 2022) and further tuned through preliminary experiments. Furthermore, the physical kinematics, control settings, task constraints, and environmental disturbances of the USVs utilized within the simulation platform are detailed in Table 3.

4. Numerical experiments and results

4.1. Performance comparison with baseline methods

To evaluate the effectiveness of the proposed approach, we compare the proposed method with three representative baselines that adopt different state representation strategies. The Grid-based method projects local observations onto a fixed-resolution 2D occupancy map and processes them with a CNN (Woo and Kim, 2020). The Fully-Connected method concatenates the features of the K nearest neighbors into a fixed-length input vector (Zhang et al., 2025b). The Pooling-based method aggregates individual entity embeddings through a symmetric pooling operation similar to DeepSets (Zaheer et al., 2017).

Algorithm 1 TMAPPO training procedure for multi-USV multi-target cooperative encirclement

Require: Max training steps $I_{\max}$, rollout horizon T, number of pursuers N, number of targets M, learning rates η_θ, η_ϕ, clip fraction ϵ, discount γ, GAE parameter λ, optimization epochs K.

Initialize shared policy network π_θ and centralized critic V_ϕ.
Initialize on-policy rollout buffer $\mathcal{D} \leftarrow \emptyset$.
for update $iter = 1$ **to** $I_{\max}/T$ **do**
  Reset environment and obtain s_0, $\{o_{i,0}, \mathbf{x}_{i,0}\}_{i=1}^{N}$.
  Initialize GRU hidden states $\mathbf{h}_0 = \{h_{i,0}\}_{i=1}^{N} \leftarrow \mathbf{0}$.
  for $t = 0$ **to** $T-1$ **do**
    for each pursuer $i \in \{1, \dots, N\}$ **do**
      Compute hidden state $h_{i,t}$ from $(o_{i,t}, \mathbf{x}_{i,t}, h_{i,t-1})$.
      Sample $a_{i,t}^{A} \sim \pi_\theta(\cdot \mid h_{i,t})$ and store $\log \pi_\theta(a_{i,t}^{A} \mid h_{i,t})$.
    end for
    for each evader $j \in \{1, \dots, M\}$ **do**
      Compute evader action $a_{j,t}^{B}$ using the predefined APF-based strategy.
    end for
    Execute joint action $\mathbf{a}_t$ and observe r_t, s_{t+1}, and $\{o_{i,t+1}, \mathbf{x}_{i,t+1}\}_{i=1}^{N}$.
    Store $\{s_t, \mathbf{a}_t, r_t, s_{t+1}\}$ in $\mathcal{D}$.
  end for
  Retain a frozen copy of the current policy, $\theta_{\text{old}} \leftarrow \theta$.
  Compute TD error δ_t, advantage estimate $\hat{A}_t$, and return target $\hat{R}_t$ from $\mathcal{D}$:

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l}, \quad \hat{R}_t = \hat{A}_t + V_\phi(s_t)$$

  for epoch $k = 1$ **to** K **do**
    for each mini-batch $\mathcal{B} \subset \mathcal{D}$ **do**
      Compute $\rho_{i,t}(\theta) = \frac{\pi_\theta(a_{i,t}^{A} \mid h_{i,t})}{\pi_{\theta_{\text{old}}}(a_{i,t}^{A} \mid h_{i,t})}$ for $(i,t) \in \mathcal{B}$.
      Update policy θ by ascending gradient:

$$\theta \leftarrow \theta + \eta_\theta \nabla_\theta \frac{1}{|\mathcal{B}|} \sum_{(i,t)\in\mathcal{B}} \min\left(\rho_{i,t}(\theta)\hat{A}_t,\ \text{clip}(\rho_{i,t}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)$$

      Update centralized critic ϕ by descending the stochastic gradient:

$$\phi \leftarrow \phi - \eta_\phi \nabla_\phi \frac{1}{|\mathcal{B}|} \sum_{(i,t)\in\mathcal{B}} \left(V_\phi(s_t) - \hat{R}_t\right)^2$$

    end for
  end for
  Clear rollout buffer $\mathcal{D} \leftarrow \emptyset$.
end for

Fig. 6 compares the learning curves of the proposed algorithm and the three baselines in a 5-pursuer, 3-evader training scenario (5-vs-3). The proposed Transformer-based approach, depicted by the red curve, shows a relatively slow but steady improvement trend. During approximately the first 10^6 training steps, its performance improves more slowly and shows larger variance than the baselines. After this stage, however, it steadily surpasses all competing methods and converges to the highest reward level. This early fluctuation is likely related to the time required for the

Table 2
Hyperparameters of the proposed algorithm

Parameter	Value
Transformer layers (L)	2
Attention heads	4
Policy embedding dimension (d_{model})	64
Policy MLP hidden dimension	256
GRU hidden dimension	64
Critic MLP hidden dimension	128
Maximum training steps ($I_{\max}$)	3×10^6
Rollout horizon (T)	256 steps
Optimization epochs (K)	10
Mini-batch size	16
Discount factor (γ)	0.99
GAE parameter (λ)	0.95
PPO clipping parameter (ϵ)	0.2
Policy learning rate (η_θ)	3e-4
Critic learning rate (η_ϕ)	5e-4
Entropy coefficient (c_{ent})	0.01
Base capture reward (r_{cap})	20
Collision safety margin (d_{coll})	5 m
Capture reward weight (k_1)	1.0
Dense reward weight (k_2)	1.0
Approach reward weight (k_3)	0.1
Collision penalty weight (k_4)	2.0

Table 3
Parameter settings of the multi-USV simulation environment

Parameter	Value
Pursuer swarm sizes (N)	$\{3, 5, 10\}$
Target swarm sizes (M)	$\{1, 2, 3, 5, 10, 20\}$
Time step (Δt)	1 s
Pursuer surge force	10 N
Evader surge force	5 N
Yaw moment action set ($\mathcal{A}$)	$\{-1, -0.75, -0.5, -0.25, 0, 0.25, 0.5, 0.75, 1\}$ N·m
Perception range (d_{sense})	150.0 m
Capture radius (d_{cap})	30 m
Minimum pursuers for capture	3
Maximum encirclement angle (α_{cap})	$\pi/2$ rad
Repulsion gain (k_{rep})	1.0
Wind disturbance speed	5.0 m/s
Current disturbance speed	0.5 m/s

multi-head attention mechanism to learn informative interaction patterns. Once the attention patterns become stable, the model more effectively filters irrelevant information and captures fine-grained pairwise interactions, leading to better encirclement performance.

In contrast, the Pooling-based method, indicated by the blue curve, displays the most stable and smooth learning trajectory, quickly reaching the second-best performance level. This stability stems from the use of aggregation mechanisms, such as mean-pooling, which simplify the optimization landscape. However, its growth plateaus noticeably in the later stages. This limitation suggests that aggressive feature compression averages out the distinct spatial relationships required for precise coordination, thereby limiting the final performance.

The Grid-based method, denoted by the green curve, shows rapid initial improvement, indicating that it can learn basic coordination strategies. However, its performance levels off early and is accompanied by significant oscillation in the final stages. This behavior suggests that the policy is limited by the loss of spatial precision. As the scenario density increases, the grid with fixed resolution becomes less effective in distinguishing precise relative positions. While increasing resolution could theoretically mitigate this issue, it would introduce prohibitive computational costs.

The Fully-Connected method, represented by the brown curve, yields the poorest performance, characterized by slow convergence and high variance throughout the training process. This weakness is mainly associated with the limited scalability of fixed-length input representations. As the input vector size grows linearly with the swarm size, the network becomes increasingly difficult to optimize effectively and is more likely to converge to poor local optima. Although such baselines may still perform adequately in small-scale tests, as also indicated by the quantitative results in Table 4, their performance degrades substantially as the scenario scale increases. This result further supports the use of the attention-based architecture in large-scale swarms.

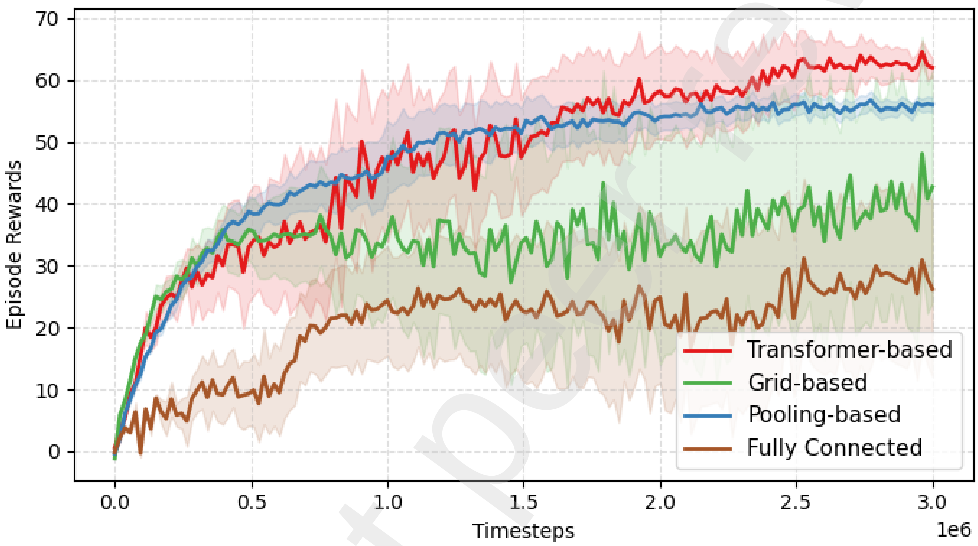

Figure 6: Training reward curves of the proposed and baseline methods under the 5-vs-3 setting

Table 4 presents a quantitative comparison of the mean reward and encirclement success rate (SR) under different training settings. In the small-scale 3-vs-1 training setting, the Fully-Connected baseline achieves the highest performance, with a success rate of 96.5% and a reward of 28.5. This indicates that fixed-length feature representations can still be effective in relatively simple settings. The proposed Transformer-based method also performs competitively, achieving a success rate of 94.5%. However, as the training setting scales from 5-vs-3 to 10-vs-5, the limitations of fixed-size input representations become evident. The Fully-Connected method shows a pronounced performance decline, with its success rate dropping to 35.0% in the 10-vs-5 setting. In contrast, the Pooling-based method still maintains success rates above 80% in the larger settings, whereas the proposed method further improves them to 96.0% and 92.5%, respectively. Moreover, the proposed Transformer-based method achieves the highest rewards in both settings. These results indicate that, although pooling-based aggregation can handle variable input sizes, the self-attention mechanism is more effective in capturing the fine-grained pairwise interactions required for precise cooperative encirclement.

4.2. Trajectory analysis of cooperative encirclement in different scenarios

Fig. 7, Fig. 8, and Fig. 9 illustrate the evolution of the cooperative encirclement process in three scenarios with increasing scale: 3-vs-1, 5-vs-3, and 10-vs-5. In these trajectory plots, the red markers labeled U^A denote the pursuers, whereas the blue markers labeled U^B denote the evaders. The shaded trajectories show the recent motion history of each agent. A captured target is marked in gray, and the dashed circle indicates the capture region of the target.

In the basic 3-vs-1 scenario, a clear cooperative pattern can be observed. At 100 s, the pursuers form a partial encirclement around the target. At approximately 140 s, when the evader attempts an abrupt heading change to escape,

Table 4
Comparison of reward and success rate under different training settings

Training Setting	Grid-based		Fully-Connected		Pooling-based		Ours (Transformer-based)	
	Rew.	SR (%)	Rew.	SR (%)	Rew.	SR (%)	**Rew.**	**SR (%)**
3 vs. 1	15.2 ± 2.1	85.0	**28.5 ± 1.8**	**96.5**	27.8 ± 1.5	90.0	22.4 ± 1.2	94.5
5 vs. 3	37.2 ± 5.5	62.0	25.5 ± 2.2	48.5	59.8 ± 1.9	88.5	**64.5 ± 1.5**	**96.0**
10 vs. 5	54.5 ± 3.0	55.0	32.1 ± 2.8	35.0	66.4 ± 2.1	81.2	**80.9 ± 1.8**	**92.5**

the two flanking pursuers adjust their headings accordingly, which reduces the available escape space. By 185 s, the target is captured within a stable triangular formation, indicating that the proposed method can achieve coordinated spatial positioning in the simplest setting.

Fig. 8 shows the swarm behavior in a 5-vs-3 scenario. Without explicit target assignment commands, the pursuers gradually separate into local sub-groups according to the target distribution and relative positions. At 350 s, the upper sub-group has already completed one capture, whereas the lower sub-group is still adjusting its formation in response to the maneuvers of the remaining evaders. By 534 s, all targets are captured, suggesting that the learned policy supports effective local coordination across multiple targets.

Fig. 9 presents the 10-vs-5 scenario. Despite the denser environment, the trajectories at 150 s remain smooth, and no obvious inter-pursuer collisions are observed. At 388 s, multiple encirclement formations are established simultaneously. Even with the increased task scale, the pursuers eventually converge on all five targets, which is consistent with the strong performance reported in the quantitative results.

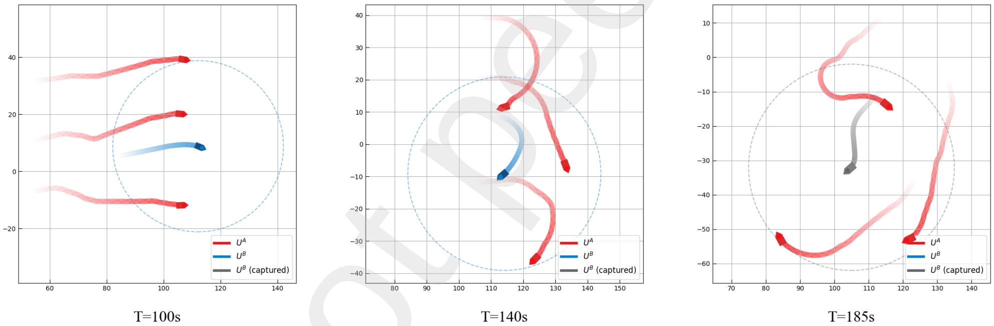

Figure 7: Trajectory visualization of the 3-vs-1 single-target encirclement scenario.

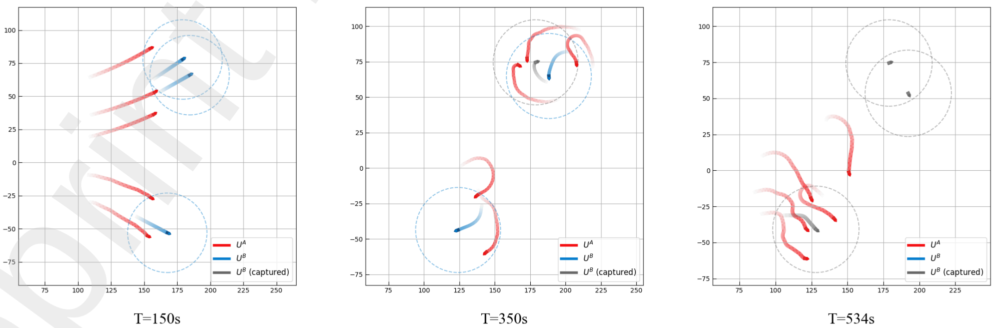

Figure 8: Trajectory visualization of the 5-vs-3 multi-target encirclement scenario.

To further examine the behavior of the learned policy under heavier task loads, additional experiments were conducted in large-scale scenarios with high evader densities. Fig. 10 and Fig. 11 illustrate the trajectories for a 10-vs-10 balanced scenario and a 10-vs-20 disadvantaged scenario, respectively.

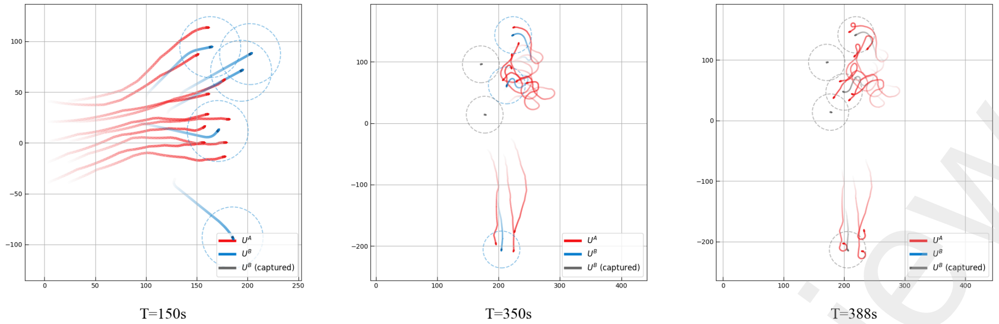

Figure 9: Trajectory visualization of the 10-vs-5 large-scale encirclement scenario.

Fig. 10 presents the results of the 10-vs-10 scenario, in which the pursuers face an equal number of evaders. At 150 s, the targets remain relatively scattered. Compared with the smaller scenarios, the higher vessel density leads to more frequent interaction among neighboring agents. By 350 s, the swarm has formed several local engagement regions around different targets. At 1000 s, most targets have been captured, indicating that the policy remains effective when the number of targets increases substantially.

Fig. 11 investigates the case of numerical inferiority, where 10 pursuers are tasked with capturing 20 evaders. In this scenario, simultaneous encirclement of all targets is not feasible. Nevertheless, the trajectories suggest a sequential capture pattern. At 150 s, the pursuers concentrate on targets that are easier to isolate instead of dispersing uniformly across the entire scene. At 500 s, the central region exhibits dense interactions, and the pursuers repeatedly complete one local capture before redirecting to nearby uncaptured targets. At 1000 s, most evaders have been captured despite the two-to-one numerical disadvantage. This result suggests that, even when resources are insufficient for full global coverage, the proposed policy can still maintain effective local coordination.

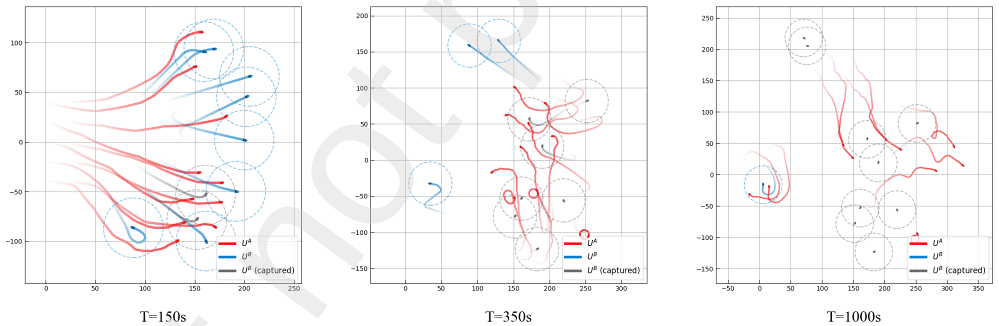

Figure 10: Trajectory visualization of the 10-vs-10 balanced scenario.

4.3. Analysis of learned attention patterns

To examine whether the proposed Transformer-based policy focuses on the most relevant agents and targets during decision-making, we visualize the attention weights of the agents. Fig. 12 illustrates the evolution of the attention vectors of pursuer A_1 in a 3-vs-2 scenario. The orange arrows denote attention links, and their widths are proportional to the corresponding weights.

At 50 s, agent A_1 is geometrically closer to target B_2. Correspondingly, a relatively high attention weight is assigned to B_2, while agents A_2 and A_3 focus on the lower target B_1. A noticeable change appears at 100 s. Although B_2 remains the closest target to A_1, the attention of A_1 shifts from the nearby B_2 toward the more distant B_1, where the sub-group $\{A_2, A_3\}$ has not yet formed a valid capture configuration. This change suggests that the policy does not allocate attention solely according to nearest-target proximity, but also focuses on agents that are important for the spatial coordination of the encirclement task. At 240 s, after B_1 has been captured, the attention assigned to that target becomes weak, and A_1, together with A_2 and A_3, redirects attention to the remaining target B_2. Noticeable attention is

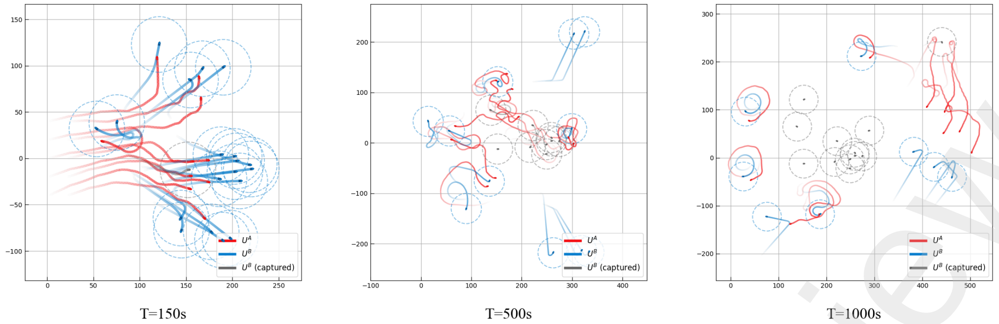

Figure 11: Trajectory visualization of the 10-vs-20 disadvantaged scenario.

also assigned to neighboring teammates, which is consistent with the need to maintain safe relative positioning during the final convergence.

To provide a quantitative view of this process, Fig. 13 visualizes the attention weight matrices. The rows represent the observing pursuers, from A_1 to A_3, and the columns denote the observed entities. At 50 s, the distribution exhibits a clear block structure: A_1 assigns the largest weight to its nearest target B_2, shown in Column 5, whereas A_2 and A_3 focus on B_1. At 100 s, the attention assigned by A_1 to B_2 decreases, while the weights on teammates A_2 and A_3 increase. This redistribution is consistent with the trajectory-based observation that A_1 is no longer responding only to the closest target. By 240 s, after B_1 is captured, its corresponding column becomes nearly inactive, indicating that the network places little attention on completed targets. All pursuers subsequently concentrate on the remaining target B_2 while preserving attention to neighboring agents.

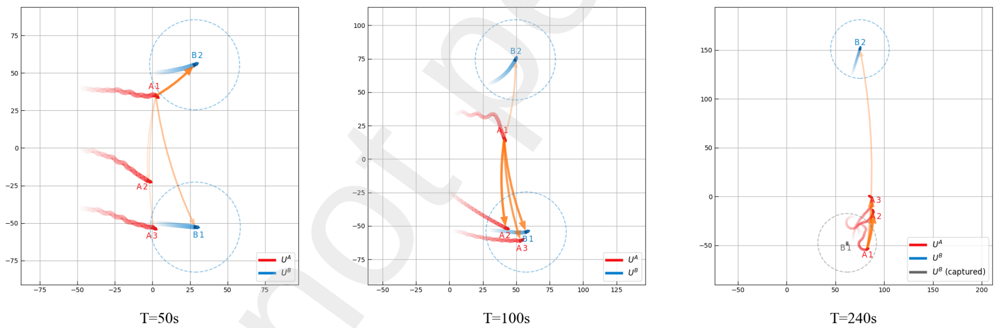

Figure 12: Visualization of the attention weights in the 3-vs-2 scenario.

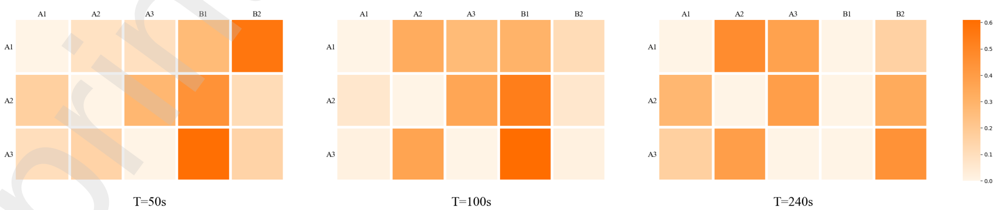

Figure 13: Temporal evolution of the attention weight matrices in the 3-vs-2 scenario.

To further investigate the adaptability of the attention mechanism in a more complex setting, Fig. 14 visualizes the evolving attention weights of agent A_5 in a 5-vs-3 scenario. At 200 s, A_5 participates in a local sub-group with teammates A_1 and A_3 to encircle target B_3, showing relatively large attention weights on these relevant entities. At 360 s, after the capture of B_3, A_5 assigns comparable attention to the two remaining targets, B_1 and B_2. At the same time, the agent maintains substantial attention on teammates A_2 and A_4, who are approaching B_1, suggesting that its decision depends on both target states and teammate positions. At 600 s, after B_1 and B_3 have been captured, the scenario reduces to single-target convergence. Agent A_5 then focuses primarily on the remaining target B_2 while continuing to attend to neighboring allies.

Fig. 15 presents the temporal evolution of the attention weight matrices, thereby providing a quantitative view of the decision process of agent A_5, which corresponds to Row 5. At 200 s, the heatmap shows a selective attention distribution in which A_5 assigns dominant weights to target B_3, shown in Column 8, and teammate A_3, shown in Column 3, indicating that the network emphasizes the entities most relevant to the local encirclement task. At 360 s, the column corresponding to the captured target B_3 decreases to near zero, while the weights assigned to the remaining targets B_1 and B_2, shown in Columns 6 and 7, become comparable. At the same time, the elevated weights on teammates A_2 and A_4, shown in Columns 2 and 4, indicate that A_5 remains sensitive to the state of the nearby teammate sub-group. By 600 s, after B_1 has been captured, A_5 shifts its dominant attention to the only remaining target B_2, shown in Column 7, while retaining off-diagonal attention on neighboring agents during the final pursuit.

These results reveal a consistent attention-allocation pattern. The learned policy does not appear to rely on a fixed nearest-target heuristic. Instead, the dominant attention shifts with the stage of the encirclement process, the status of different targets, and the relative configuration of neighboring teammates. In particular, completed targets are gradually suppressed, whereas active targets and nearby cooperating agents continue to receive higher attention. These observations suggest that the attention mechanism helps support dynamic target selection and local spatial coordination in multi-target cooperative encirclement.

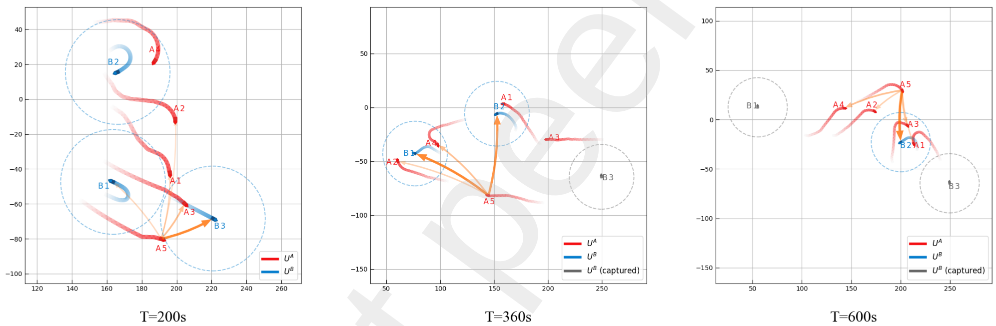

Figure 14: Visualization of the attention weights in the 5-vs-3 scenario.

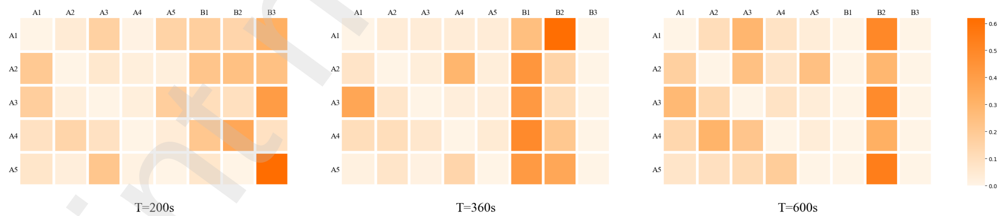

Figure 15: Temporal evolution of the attention weight matrices in the 5-vs-3 scenario.

5. Conclusion

In this paper, we proposed a Transformer-enhanced multi-agent proximal policy optimization framework for the multi-USV multi-target encirclement task. First, we developed a policy network that combines a Transformer encoder with a GRU-based recurrent module, enabling each agent to process variable-length observations, focus on critical spatial information, and enhance temporal decision-making capability. Second, we designed a comprehensive reward function that jointly considers target pursuit, collision avoidance, and encirclement uniformity, thereby encouraging safe and effective cooperative encirclement behaviors. Third, we constructed a more realistic training environment by incorporating USV dynamic characteristics and environmental disturbances, which improves the robustness and practical relevance of the learned policy.

Numerical experiments demonstrated that the proposed method achieves robust performance across multiple swarm scales. The proposed method achieved the best overall performance in the more challenging large-scale scenarios, attaining mean rewards of 64.5 and 80.9 and success rates of 96.0% and 92.5%, respectively. These results indicate that the proposed architecture scales more effectively as the numbers of agents and targets increase. The trajectory and attention analyses further show that the learned policy supports stable cooperative encirclement across scenarios of increasing complexity. The pursuers progressively form effective encirclement configurations while maintaining smooth motion and collision-aware coordination, and they remain capable of local grouping and sequential target capture in more demanding large-scale scenarios. In addition, the attention distributions vary with task progress, target status, and neighboring teammate configuration, suggesting that the proposed method supports dynamic target selection, local spatial coordination, and interpretable cooperative decision-making in multi-target encirclement tasks.

Despite the encouraging results, several limitations remain. The current validation is restricted to high-fidelity simulations, and real-world deployment is still challenged by communication delays, sensor noise, and modeling uncertainty. In addition, the present framework mainly considers a homogeneous pursuer swarm. Future work will therefore focus on sim-to-real transfer through experiments with physical USV platforms and on extending the proposed method to heterogeneous multi-agent systems with diverse sensing, actuation, and decision-making capabilities.

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